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Accounting for surface roughness in modeling of ice formation

N. G. Galanov[1,2] · A. S. Kozelkov[1,2,3] · D. Yu. Strelets[3] · E. S. Tyatyushkina[1,2]

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Abstract
This article presents a physics-mathematical model for the numerical simulation of ice formation on aircraft surfaces under standard atmospheric conditions, incorporating surface roughness arising during icing processes. The resulting roughness is described using an equivalent sand-grain roughness model and its modifications. Two modifications to the equivalent sand-grain roughness model are presented, which automatically calculate the height of the roughness based on inflow air parameters and water droplet characteristics. These models are applied in the numerical solution of airflow around the NACA0012 wing profile, both with and without accounting for surface roughness. The results of this work allow for assessing the applicability of the roughness model across a range of inflow temperatures. Results demonstrate that incorporating roughness leads to the formation of ice accretions with similar shapes and locations as observed in experiments. Overall, accounting for surface roughness in numerical icing simulations of aircraft enables the prediction of ice morphologies that more closely align with experimental data compared to results obtained without considering roughness.

Keywords Aerodynamics · Numerical simulation · Navier–Stokes equations · Logos aero-hydro software · Icing · Sand-grain roughness model

1 Introduction

Operational experience with aircraft demonstrates that icing represents one of the most dangerous natural phenomena, posing a significant safety challenge for civil aviation worldwide. Ice accretion on aircraft surfaces can substantially alter aerodynamic characteristics of wings, fuselages, and other components, leading to reduced lift generation and increased drag. In extreme cases, icing can result in catastrophic consequences, including loss of control and aircraft descent. To enhance flight safety, aviation certification standards are implemented during the design of new aircraft and for the modernization of existing airframes. Prior to operational use, any aircraft must undergo certification tests to meet regulatory agency requirements. Certification is a costly, labor-intensive, and strictly formalized process. Primary certification requirements are established by national aviation authorities such as (Rosaviation) in Russia, the European Aviation Safety Agency (EASA) in Europe, and the Federal Aviation Administration (FAA) in the United States. Russian aviation regulations [1] and certification requirements outlined in Appendices O to European (CS25) [2] and U.S. Standards for Aircraft Engine Overhaul (FAR29) [3] include sections such as “Ice protection of intake systems” and “Ice protection”. Current aircraft certification procedures for flights under icing conditions necessitate aerodynamic model testing in wind tunnels and flight tests of a real aircraft with wing ice simulators [4]. The development of numerical modeling has partially replaced physical testing with computer simulations [5, 6]. Specifically, numerical methods for calculating ice accretion have been successfully applied, now enabling the assessment of ice behavior and formation processes using computational models. In solutions presented Kozelkov et al. [5], it was determined that a more complex model incorporating surface roughness is required to better match experimental data. Rotich and Kollár [7] systematizes and presents a

✉ D. Yu. Strelets
maksmai33@gmail.com

1 Russian Federal Nuclear Center - All-Russian Research Institute of Experimental Physics, Nizhny Novgorod Region, Sarov, Russia

2 Nizhny Novgorod State Technical University N.a. R.E. Alekseev, Nizhny Novgorod, Russia

3 Moscow Aviation Institute (National Research University), Moscow, Russia

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comprehensive range of research by various authors [8–11] focused on experimental studies and numerical modeling of the influence of surface roughness on the aerodynamics and icing of investigated aircraft. For example, Gao et al. [8] demonstrated the impact of wing surface roughness on flight characteristics. An experimental installation utilized a rough aerodynamic surface of the DU91-W2-250 wing model. These findings indicate that surface roughness leads to a noticeable reduction in lift generated by the wing. This data is further corroborated by the work of other researchers Xin Guan et al. [9], who emphasize the importance of surface roughness on icing processes. Specifically, surface roughness significantly influences water droplet adhesion and ice distribution. Smoother surfaces are generally less susceptible to large-scale ice accumulation because water droplets more easily slide off the surface before freezing. Rougher surfaces can delay moisture buildup and accelerate icing [11–13]. As surface roughness increases, vortical flows near the surface may be generated at lower inflow velocities. The flow velocity in the boundary layer near a rough surface differs from that near a smooth surface [12, 14]. Numerical models are verified based on experimental studies examining the influence of surface roughness on aerodynamic properties of aircraft [4, 15–18]. Numerical methods for modeling surface roughness can be incorporated into software after validation [9, 12, 19, 20].

The most widely utilized roughness model in numerical icing simulations is the Equivalent Sand-Grain Roughness (ESGR) model [12, 20]. This model is predicated on an equivalent sand-grain height, reflecting primarily the aerodynamic effects of surface texture rather than the actual physical dimensions of the roughness elements. This parameter substitutes for the complex structure of rough surfaces with simplified mathematical formulations; consequently, the influence of surface roughness on aerodynamics and heat transfer can be predicted without requiring a detailed description of each individual roughness feature. Furthermore, this model is applicable to diverse types of rough surfaces, including those formed during icing processes. This significantly streamlines the mathematical representation of the problem and reduces computational costs. The sand-grain roughness model is commonly coupled with the Spalart–Allmaras (SA) turbulence model or a more sophisticated turbulence model such as k-ω SST (Shear Stress Transport) [23]. However, employing the ESGR sand-grain model in conjunction with the k-ω SST turbulence model enhances the prediction of turbulent flow behavior in regions characterized by significant velocity and pressure gradients–conditions frequently encountered during deformation of computational models subjected to icing. This assertion stems from the fact that the k-ω SST model exhibits greater accuracy and reliability across a broader range of flows (including adverse pressure gradient flows on aerodynamic surfaces, supersonic compression waves) compared to the standard SA turbulence model [17].

This work presents a numerical icing model incorporating surface roughness. The icing simulation process is divided into three distinct stages. In the first stage, flow motion within a viscous, compressible gas is modeled using the Reynolds-averaged Navier–Stokes equations. To account for turbulent flows, the k-ω SST (Shear Stress Transport) turbulence model is employed, supplemented with appropriate surface roughness correlations. Simultaneously, at this stage, droplet movement within a continuous fluid medium is simulated via Euler equations, assuming that droplets have negligible influence on the gas phase. The second stage focuses on modeling the motion of a liquid film along a solid surface. Finally, in the third stage, deformation of the computational model's surface occurs at locations where ice formation is predicted. A key characteristic of the numerical discretization is its implementation using the finite volume method (FVM) on arbitrary unstructured grids [21]. This described numerical model is applied to investigate aerodynamic problems for both smooth and rough profiles of the NACA0012 airfoil, with and without accounting for surface roughness. The resulting data are compared against experimental findings.

2 Mathematical model for numerical simulation of icing

2.1 Equations for calculating the aerodynamic and turbulence characteristics

The Reynolds-averaged Navier–Stokes (RANS) equations [22] are used for numerical modeling of aerodynamic flow characteristics in the computational domain. In their conservative form, in Cartesian coordinates, the equations are as follows (averaging signs omitted):

$$\begin{cases} \dfrac{\partial \rho}{\partial t} + \nabla\left(\rho \vec{u}\right) = 0, \\ \dfrac{\partial\left(\rho \vec{u}\right)}{\partial t} + \nabla\left(\rho \vec{u}\vec{u}\right) = -\nabla p + \nabla\left(\tau_{\mu} + \tau_{t}\right), \\ \dfrac{\partial\left(\rho E\right)}{\partial t} + \nabla\left(\rho \vec{u} h\right) = \nabla\left[\vec{u}\left(\tau_{\mu} + \tau_{t}\right) - \left(\vec{q}_{\mu} + \vec{q}_{t}\right)\right]. \end{cases} \quad (1)$$

In (1): ρ=density; $\vec{u}$=velocity vector with the components u, v, w; p=pressure; $E = C_v T + 0.5\left(u^2 + v^2 + w^2\right)$=the total gas energy; $H = C_p T + 0.5(u^2 + v^2 + w^2)$=the total gas enthalpy; τ_{μ} and τ_t=the molecular and turbulent components of the tangential stress tensor, respectively; q_{μ} and q_t=the molecular and turbulent components

of the heat flux density, respectively; T=temperature; $C_v = (C_pT - R/m)$=specific heat capacity at constant volume; C_p=specific heat capacity at constant pressure; R=the gas constant; m=the molar mass of the gas.

System (1) is supplemented with different types of boundary conditions: free boundary, outflow, symmetry, periodicity, and wall conditions. A detailed description of the boundary conditions can be found in [23]. To solve the Navier–Stokes equations numerically, we use the finite volume method (FVM) [24].

The system of Eq. (1) is not closed due to the unknown connection of one of the main variables of this system, τ_t and q_t, and the average flow parameters. To close the system of equations, the Boussinesq hypothesis is used, according to which a linear relationship is established between the gradients of the averaged velocity by the introduction of turbulent viscosity the introduction of turbulent viscosity μt, which is calculated using various turbulence models. In this study, the standard k − ω SST model presented in the classic form in [25].

$$\begin{aligned}\frac{\partial \rho k}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j k) &= \frac{\partial}{\partial x_j}\left[(\mu + \sigma_k \mu_t)\frac{\partial k}{\partial x_j}\right] + P_k - \beta^* \rho \omega k,\\ \frac{\partial \rho \omega}{\partial t} + \frac{\partial}{\partial x_j}(\rho u_j \omega) &= \frac{\partial}{\partial x_j}\left[(\mu + \sigma_k \mu t)\frac{\partial \omega}{\partial x_j}\right] + \gamma \frac{\rho}{\mu_t} P_k - \beta \rho \omega^2 + (1 - F_1) D_{k\omega},\end{aligned} \tag{2}$$

where ω=ε/k=the specific rate of kinetic energy dissipation, P_k=the generational member, $\mu_t = k/\omega$=the turbulent viscosity, F_1=an empirical function.

The empirical constants of the model are determined through the corresponding constants $k-\omega$ of the models by using F_1:

$$\begin{aligned}\sigma_k &= F_1 \sigma_{k1} + (1 - F_1)\,\sigma_{k2},\\ \sigma_\omega &= F_1 \sigma_{\omega 1} + (1 - F_1)\,\sigma_{\omega 2},\\ \beta &= F_1 \beta_1 + (1 - F_1)\,\beta_2,\end{aligned} \tag{3}$$

$$\begin{aligned}F_1 &= \tanh\left(\arg_1^4\right),\\ \arg_1 &= \min\left[\max\left(\frac{\sqrt{k}}{0.09\omega d_w}, \frac{500v}{d_w^2 \omega}\right), \frac{4\rho\sigma_{\omega 2} k}{CD_{k\omega} d_w^2}\right],\\ CD_{k\omega} &= \max\left\{D_{k\omega}, 10^{-20}\right\}\end{aligned} \tag{4}$$

Subscripts «1» and «2» in (3) are related, respectively, to the constants of the $k-\omega$ model

$$\begin{aligned}&\sigma_{k1} = 0.85,\ \ \sigma_{\omega 1} = 0.5,\ \ \ \ \beta_1 = 0.075,\\ &\sigma_{k2} = 1.0,\ \ \sigma_{\omega 2} = 0.856,\ \ \beta_2 = 0.0828,\\ &\beta^* = 0.09,\ \ k = 0.41,\ \ a_1 = 0.31,\ \ \gamma = \beta/\beta^* - \sigma_\omega k^2/\sqrt{\beta^*}.\end{aligned} \tag{5}$$

d_w=the distance to the nearest point on a solid surface.

2.2 The logarithmic law of wall function and the sand-grain roughness model (ESGR)

Experimental and numerical studies have shown differences in flow structure and behavior on rough and smooth surfaces [4, 9, 12, 17–20]. To better understand this roughness modeling technology, we will examine the flow structure near a solid surface, schematically illustrated in Fig. 1.

At the solid surface, flow velocity is zero. The influence of the wall diminishes progressively with increasing distance from the surface. The turbulent boundary layer is typically divided into two distinct layers: the outer and inner layers. The outer layer is characterized by dominant turbulent effects. The inner layer is further subdivided into three sublayers, each exhibiting different flow characteristics:

- Viscous sublayer—this sublayer is almost laminar, in contact with the wall surface, which is why viscous effects prevail in it. The average flow velocity depends on the flow density, viscosity, distance to the wall and the shear stress on the wall.
- Buffer sublayer—this sublayer is a transitional layer between a viscous sublayer and a logarithmic one.
- Logarithmic sublayer—in this sublayer, viscous and turbulent effects equally affect the flow.

Wall functions are used to calculate the flow velocity in the turbulent boundary layer when modeling the flow for high Reynolds numbers [25]. To provide stability of counting after deformation of design grid model, which occurs during formation of ice growths, dimensionless distance to wall (y^+) must be more than 30. The y^+ was derived as

$$y^+ = \frac{y u_\tau}{v} \tag{6}$$

where y is first layer thickness, u_τ is friction velocity, and $\nu = \mu/\rho$ is kinematic viscosity.

The dimensionless velocity in the boundary layer is calculated using the following formula:

$$u^+ = \frac{1}{\kappa}\ln(Ey^+) + B \tag{7}$$

where $B = 5$ is constant value,$\kappa = 0.41$ is Karman constant, E is displacement magnitude.

The roughness value complements Eq. (8) as follows [14]:

$$u^+ = \frac{1}{\kappa}\ln(Ey^+) + B - \Delta B \tag{8}$$

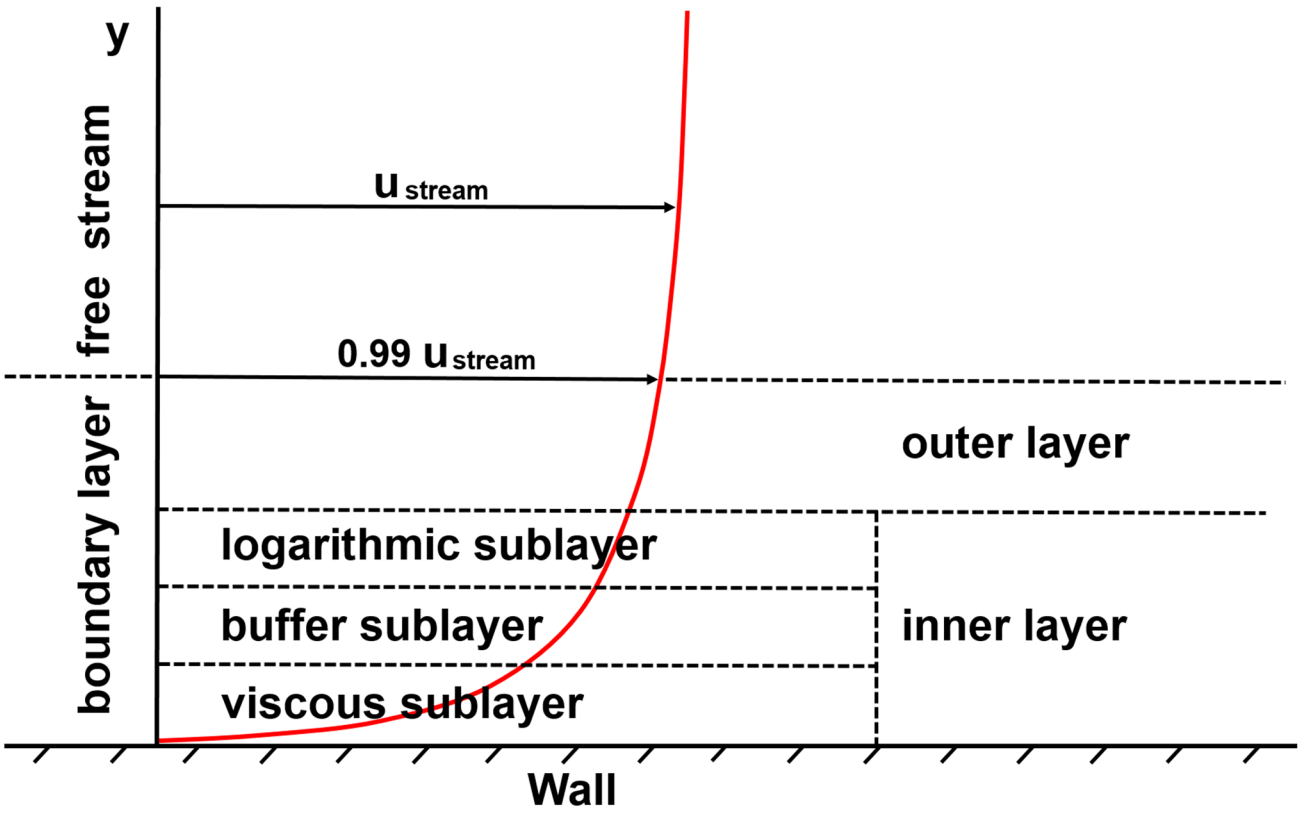


Fig. 1 Flow structure in turbulent boundary layer

$$\Delta B = \frac{1}{\kappa}\ln(f) - 3,5;\ B = 5 \tag{9}$$

$$u^+ = \frac{1}{k}\ln(E\prime y^+) + 8,5 \tag{10}$$

where $E\prime$ is equal to E/f, and f is obtained based of the value of k_s^+:

$$f = \begin{cases} 1\, k_s^+ < 2.25 \\ \left[\left(\dfrac{k_s^+ - 2.25}{87.75}\right) + Ck_s^+\right]^a \quad 2.25 \le k_s^+ < 90 \\ B + Ck_s^+\, k_s^+ \ge 90 \end{cases} \tag{11}$$

The rest of the coefficients is obtained as follows: $a = \sin\frac{\pi}{2}\frac{\ln(k_s^+/2.25)}{\ln(90/2.25)}$, $B = 0$, $\mathrm{C} = 0.253$.

The roughness k_s^+ is used to calculate the function (11).

$$k_s^+ = k_s u_\tau/\nu \tag{12}$$

where k_s=the equivalent sand-grain roughness height in meters, $\nu = \mu/\rho$=kinematic viscosity, μ=molecular viscosity, ρ=density, u_τ=the friction velocity.

In this case, the equivalent sand-grain roughness height k_s is set by the user when preparing the task parameters for calculation.

Figure 2 shows the distribution of u^+ in the three sublayers of the inner turbulent boundary layer. Figure 2 also shows the change in the dimensionless flow velocity when using an additional term in the wall function that takes into account the magnitude of roughness. Increasing the sand-grain roughness height k_s^+ increases the value of ΔB, which in turn decreases the dimensionless flow rate u^+ in the boundary layer.

To account for roughness in the SST turbulence model, Knopp et al. [20] proposed to calculate the turbulent kinetic energy k and the turbulent specific velocity of dissipation ω on the wall, for the system of Eq. (2), as a function of equivalent sand-grain roughness height k_s^+:

$$\omega = \min\left(\frac{u_\tau}{\sqrt{\beta}\kappa d_0}, \frac{60\nu}{\beta y^2}\right),\ k = \min\left(1, \frac{k_s^+}{90}\right)\frac{u_\tau}{\sqrt{\beta}} \tag{13}$$

$$d_0 = 0.03 \cdot k_s \cdot \min\left(1, \left(\frac{k_s^+}{30}\right)^{\frac{2}{3}}\right) \cdot \min\left(1, \left(\frac{k_s^+}{45}\right)^{\frac{1}{4}}\right) \cdot \min\left(1, \left(\frac{k_s^+}{60}\right)^{\frac{1}{4}}\right) \tag{14}$$

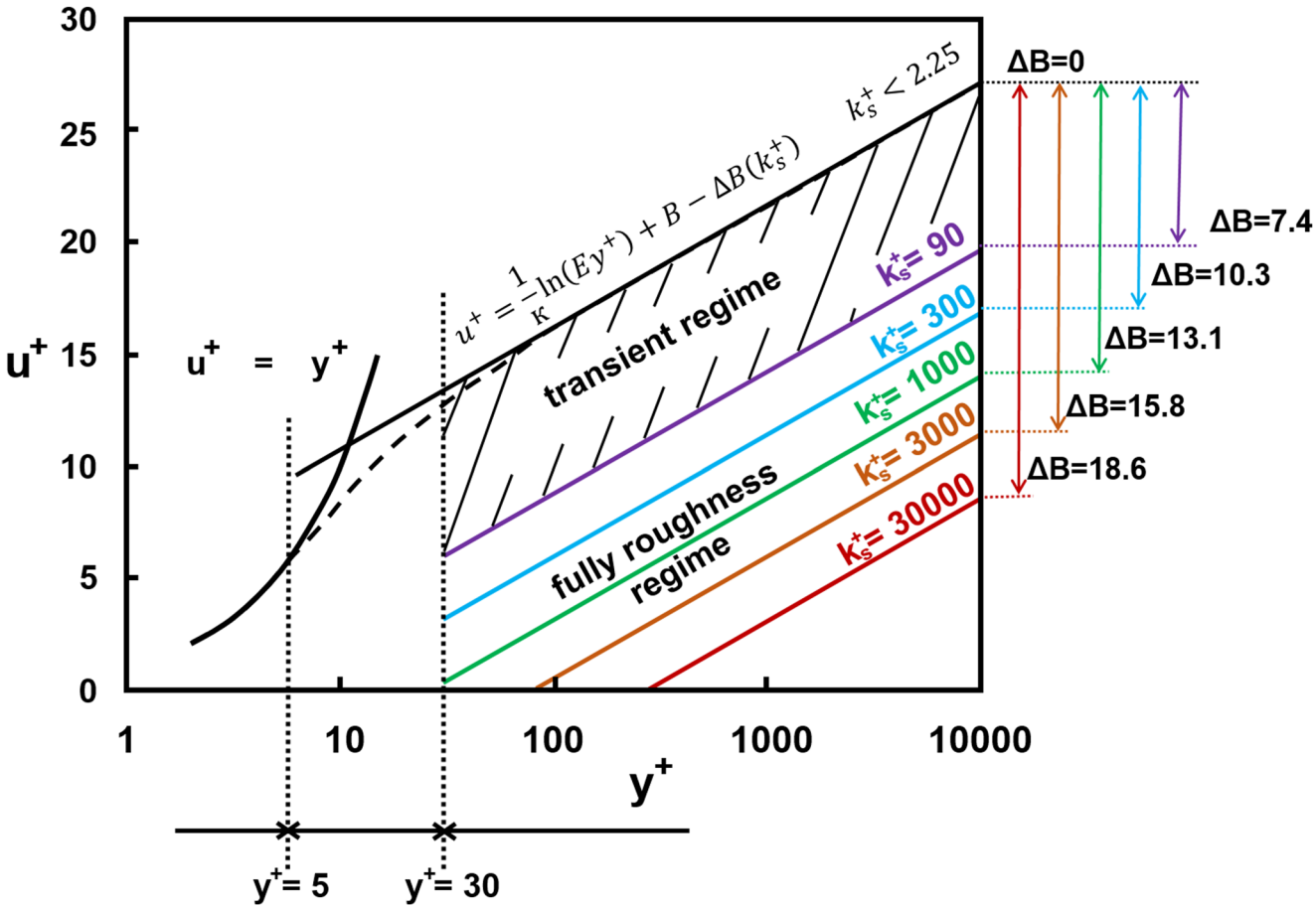


Fig. 2 Dimensionless flow velocity as a function of dimensionless distance to the wall

The proposed model was calibrated in [26] and showed good agreement with a completely rough surface [27].

This roughness model (ESGR) has been described in various papers [4, 12]. The roughness type is determined by roughness magnitude k_s^+ [28]:

- For a completely smooth surface, $k_s^+ < 2.25$;
- For transitionally rough surfaces, $2.25 \leq k_s^+ < 90$;
- For completely rough surfaces, $k_s^+ \geq 90$.

Figure 3 schematically shows the surface roughness of a relatively viscous sublayer and how the equivalent sand-grain roughness is related to the views presented.

Knopp et al. [20], Orych et al. [27], Cebeci and Bradshaw [28], it is shown that when using the above model of equivalent sand-grain roughness, the obtained aerodynamic characteristics are in better agreement with experimental data; however, the presented modification of the logarithmic wall function is implemented for a "conditionally" smooth surface without taking into account the growing ice. In experimental observations, there is an inhomogeneity in the roughness of the ice formed on the surface of the airfoil [29], which affects its aerodynamic properties. To account for this heterogeneity, special corrections are introduced that take into account the extremely large surface roughness of the formed ice. Two roughness correction options have been integrated specifically for icing cases: NASA [30] and Shin [31]. In the case of a NASA correction, the equivalent sand-grain roughness height is determined by the following set of equations:

$$k_s = \left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{LWC} \cdot \left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{T_\infty} \cdot \left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{V_\infty} \cdot (k_s/c)_{base} \cdot c \tag{15}$$

$$\left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{LWC} = 0.5714 + 0.2457 \cdot LWC + 1.2571 \cdot (LWC)^2 \tag{16}$$

$$\left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{T_\infty} = 0.047 \cdot T_\infty - 11.27 \tag{17}$$

$$\left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{V_\infty} = 0.4286 + 0.0044139 \cdot V_\infty \tag{18}$$

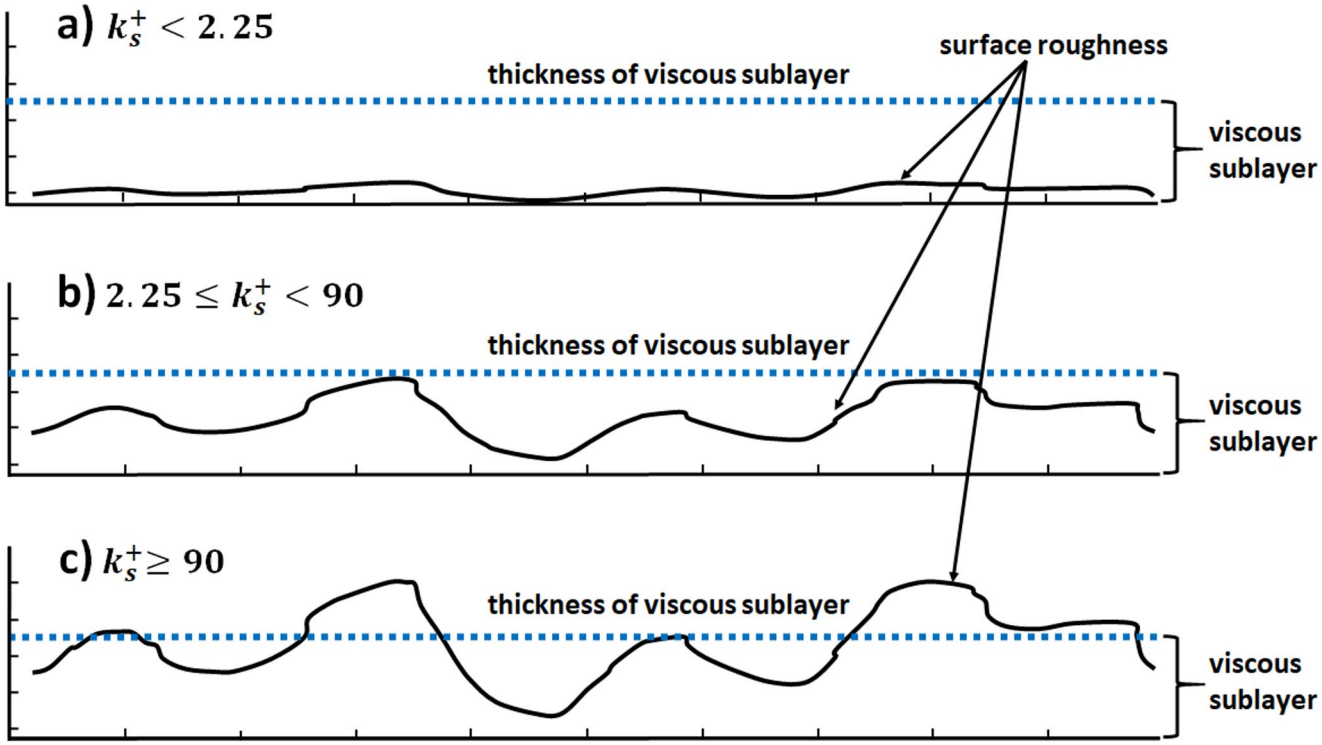


Fig. 3 Surface roughness relative to the viscous sublayer: **a** hydraulically smooth; **b** transitionally rough; **c** completely rough

where k_s = equivalent sand-grain roughness (ESGR) height in m, c = the chord length of the airfoil in m, $(k_s/c)_{base} = 0.001177$, T_∞ = the freestream temperature in K, V_∞ = the freestream velocity in m/s, LWC = cloud liquid water content in g/m^3.

In the Shin correction, the equivalent sand-grain roughness height does not depend on the freestream velocity, but depends on the average volume diameter of the droplets and has the following form:

$$k_s = 1.3678 \cdot \left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{LWC} \cdot \left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{T_\infty} \cdot \left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{MVD} \cdot (k_s/c)_{base} \cdot c \tag{19}$$

$$\left[\frac{k_s/c}{(k_s/c)_{base}}\right]_{MVD} = \begin{cases} 1, & MVD \le 20 \\ 1.667 - 0.0333 \cdot MVD, & MVD > 20 \end{cases} \tag{20}$$

where MVD = the average volume diameter of the droplets in μm.

The main difference between the above corrections for calculating the ice roughness through NASA and Shin correlations and the resulting value from Eq. (11) is the method for calculating the roughness value. Using Eq. (11), you need to know the roughness height accurately depending on the problem, while in the NASA model, the equivalent sand-grain roughness are calculated using the water content of the stream (LWC), temperature (T) and the speed of the incoming stream (V), and in the Shin model, the equivalent sand-grain roughness are calculated using the water content of the stream (LWC), the temperature of the incoming stream (T) and mean droplet diameter (MVD). In this case, the equivalent sand-grain roughness height is calculated automatically depending on the specified parameters of the incoming flow.

It should be noted that there are limitations for the presented correlations of Shin and NASA: when the average droplet volume (MVD) is greater than 50 μm or the ambient flow temperature is below 240°K, the results obtained by formulas (17) and (20) will be negative. In numerical calculations, negative values of the equivalent roughness value are automatically replaced with zero values. If you consider modifying the Shin and NASA correlations for wider use, you should consider the following aspects:

1) At temperatures below 240 °K, water droplets in the environment are not in liquid form, but in a crystallized state. In this case, the above physical and mathematical model should be applied taking into account the fact that there is no water film on the surface of the body. In the absence of a water film on the surface, roughness will only affect the flow rate in the vicinity of the body. This influence is small and, as will be shown below, practically does not affect the shapes of ice growths formed at the leading edge of the wing.

2) The limitation of the average droplet volume (*MVD*) of 50 μm is due to the mechanism of modeling Supercooled Large Droplets (SLD). It is generally accepted that SLD (Small Liquid Droplet) droplets possess diameters exceeding 50 μm, necessitating the inclusion of droplet fragmentation and spraying phenomena upon contact with surfaces [32]. The resulting ice surface exhibits a more complex morphology and increased roughness height compared to values calculated by the presented roughness model. Consequently, this roughness model is not applicable to SLD droplets.

For the above parameters exceeding the limits, the ESGR height will be negative. For clarity, Fig. 4 shows the equivalent sand-grain roughness height distribution calculated using the Shin correlation as a function of the freestream temperature (*T*) and the average diameter of the droplets (*MVD*). For this distribution, a freestream water content (*LWC*) of 0.001 kg/m^3 and a chord length of 1 m were used.

2.3 Equations for motion of liquid droplet as a continuous medium in a gas flow

The calculated fields obtained in the previous step are used in numerical modeling of the motion of water droplets. To numerically model the motion of liquid droplets in the calculated area in the form of a continuous medium, an Eulerian approach is used, in which the distributions of gas-dynamic fields are taken as input data. This calculation is performed after each iteration of solving the system of Eq. (1) by solving the system of water volume fraction, particle velocity, and temperature equations [33].

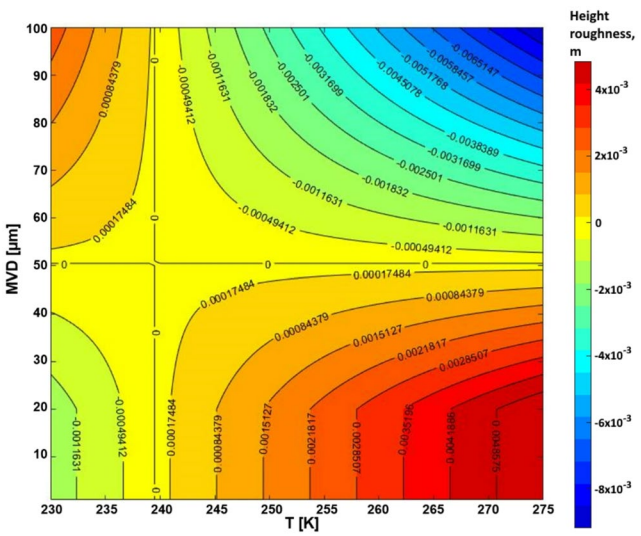


Fig. 4 Equivalent sand-grain roughness height vs. temperature freestream (T) and droplet diameter (MVD)

$$\begin{cases} \frac{\partial \alpha}{\partial t} + \nabla \cdot (\alpha \vec{u}_w) = 0 \\ \frac{\partial \alpha \vec{u}_w}{\partial t} + \vec{u}_w \nabla (\alpha \vec{u}_w) = \frac{C_D \mathrm{Re}_d}{24K} (\vec{u}_a - \vec{u}_w) + (1 - \frac{\rho_a}{\rho_w}) \frac{1}{Fr^2} g \\ \frac{\partial \alpha T_w}{\partial t} + \nabla \cdot (\alpha \vec{u}_w T_w) = 0 \end{cases} \quad (21)$$

where α is defined as the volume fraction of water, $\vec{u}_w$ is the velocity of droplets, $\vec{u}_a$ is the carrier phase velocity (the aerodynamic flow is determined from Eq. (1)), C_D is a drag coefficient, Re_d is the droplets Reynolds number, ρ_a is density of air, ρ_w is density of the liquid phase, $Fr = U_\infty / \sqrt{Lg}$ is the Froude number; U_∞ is the speed of air at freestream; *L* is the characteristic length (typically the airfoil chord length); $K = \rho d^2 U_\infty / 18 L \mu$ is an inertia parameter; μ is the dynamic viscosity of air, and T_w is the temperature of the continuum phase, g is gravity vector.

Due to adiabatic compression and an increase in flow density in the region of the stagnation point, the gas temperature increases. However, due to the limited heat transfer rate, the temperature of the water droplets may not correspond to the ambient temperature at high water content or at high ram-stream speeds. This physical effect must be taken into account when numerical modeling of icing at supersonic speeds. To do this, the system of equations of motion of liquid droplets uses the temperature transfer equation.

This system of Eq. (2) must be supplemented by boundary conditions. In general, we use three main types of boundary conditions for the liquid phase: inflow boundary conditions, outflow boundary conditions, and impermeable boundary. For the inflow and outflow boundaries, a boundary condition of the first kind is used, for which parameter values are set as constants, functions, and tables. In the icing problem, such parame-ters include the velocity and temperature of the carrier phase, the volume fraction and diameter of water droplets.

Ice accretion occurs on those areas of the surface where liquid droplets fall in sufficient quantities and a water film is formed. To determine the parts of the surface most susceptible to the impingement of droplets, the collection efficiency of liquid droplets up-on collision with the surface is introduced. This parameter characterizes the ability of the considered surface area to collect water and is defined as the local mass flux of water onto the airfoil surface normalized by the product of the velocity and volume fraction of water in the liquid phase at the inflow boundary [33]:

$$\beta = \frac{\alpha \rho u_i}{(LWC)\, U_\infty} \frac{S_i}{|S_i|} \quad (22)$$

where α α is the volume fraction of droplets, *LWC* is liquid water content, U_∞ is speed of air at free stream, S_i is the local area normal.

2.4 Equations for water film motion over solid surface and compatibility relations

To calculate the collection efficiency of the liquid phase (the forming water film) on a solid surface, we calculate the mass, momentum and temperature of the droplets, which are used in the equation of motion of the liquid film, called the Shallow-Water Icing Model (SWIM) [34] and written as follows:

$$\begin{cases} \rho_w \left[\frac{\partial h_f}{\partial t} + \operatorname{div}\left(\bar{u}_f h_f\right) \right] = \dot{m}_d - \dot{m}_{evap} - \dot{m}_{ice}, \\ \rho_w \left[\frac{\partial h_f c_{p,w} \tilde{T}}{\partial t} + div\left(\bar{u}_f h_f c_{p,w} \tilde{T}\right) \right] = \\ = \left[c_{p,w} \tilde{T}_d + \frac{|\vec{u}_d|^2}{2} \right] \times U_\infty LWC\beta - 0.5\left(L_{evap} + L_{sub}\right)\dot{m}_{evap} \\ + \left(L_{fus} - c_{p,ice}\tilde{T}\right)\dot{m}_{ice} + \dot{Q}_{conv} + \dot{Q}_{cond} \end{cases} \tag{23}$$

where ρ_w = the water density, h_f = film thickness, $\overline{u}_f$ = the speed of movement of the film in the direction x, $\dot{m}_d$ = the mass flow entering the control volume as a result of falling droplets, $\dot{m}_{evap}$ = evaporation/sublimation flow, $\dot{m}_{ice}$ = the mass flow leaving the control volume as a result of freezing, $c_{p,w}$ = specific heat of water, $\tilde{T}$ = scaled temperature ($= T - T_{ref}$), $\tilde{T}_d$ = scaled temperature of droplets, $\vec{u}_d$ = impact droplet velocity, U_∞ = velocity of air at infinity, LWC = cloud liquid water content, L_{evap} = evaporation latent heat of water, L_{sub} = sublimation latent heat of ice, L_{fus} = fusion latent heat of ice, $\dot{Q}_{conv}$ = convective heat flux from air, $\dot{Q}_{cond}$ = conductive heat flux from structure.

There are three unknown parameters in the system of Eq. (4): the film thickness h_f, the equilibrium temperature $\tilde{T}$ of the air/film/ice/wall interface, and the icing mass flux $\dot{m}_{ice}$, so additional compatibility relations are needed to close this system. The way the compatibility relations are written based on the experimental observations in [37] is the following:

$$h_f \geq 0,\ \dot{m}_{ice} \geq 0,\ h_f \tilde{T} \geq 0,\ \dot{m}_{ice}\tilde{T} \leq 0 \tag{24}$$

In accordance with (24), the systems of nonlinear Eq. (23) is divided into three modifications:

Variant I, $\dot{m}_{ice} = 0,\ h_f > 0, \tilde{T} > 0$;

Variant II, $\tilde{T} = 0,\ T = T_0,\ \dot{m}_{ice} > 0,\ h_f > 0$;

Variant III, $h_f = 0,\ \tilde{T} < 0,\ \dot{m}_{ice} > 0$.

For each of the options, a different system of nonlinear equations is used, sampled according to an explicit scheme.

2.5 Model surface motions

The next step is to calculate the ice formed on the surfaces of the models. Based on the mass of the ice buildup, using motion algorithms [35], new coordinates of nodes on the surface of the calculated model are determined, after which it is deformed. Algorithms of motion are based on the IDW (Inverse Distance Weighting) method [38], which refers to methods for direct interpolation of control node displacements. This method is based on an interpolation function of the following form:

$$s(x) = \frac{\sum w_i(x) s_j(x)}{\sum w_j(x)} \tag{25}$$

$$w_j(x) = A_j \left[\left(\frac{L}{\left\| x - x_{b_j} \right\|} \right)^a + \left(\frac{\alpha L}{\left\| x - x_{b_j} \right\|} \right)^b \right], \tag{26}$$

where $w_i(x)$- there is a weight function, $s_j(x)$ it is determined not only by the displacement of the j-th boundary node, but also by the position of the point x for which the displacement is calculated, A_j—node weight proportional to the area of its surrounding faces, L is the characteristic area size, αL—the scale of the boundary region, a and b are some exponents.

A more detailed description of the method is presented in [35, 36].

3 Results and discussion

3.1 Analysis of the applicability of the ESGR model in solving aerodynamic problems

To quantitatively assess the influence of ESGR on ice formation, an investigation into the aerodynamic effects of surface roughness on aircraft without icing is warranted. Consequently, we will be performed numerical studies examining airflow around both the smooth and rough NACA0012 airfoils, characterized by a chord length of 1 m. The NACA0012 airfoil was taken from open sources [37]. To determine the independence of the results from the mesh size, computational domains with the following number of cells were selected: geom1) 449 × 129, geom2) 897 × 257, and geom3) 1793 × 513. The number of cells along the Oz axis is 1 cell. The following freestream

parameters were used in numerical calculations [38]: Mach number: $M = 0.15$, Reynolds number: $\mathrm{Re} = 1.25 \times 10^6$, temperature: $T = 300K$. We used the coupled solver type, the implicit difference scheme, the AUSMPW+convective flow calculation scheme, and the SST turbulence model. As a result, the NACA0012 lift curves were plotted agianst the test data, as well as the results of the SU2, WMB program codes [39] for the following angles of attack: 0, 5, 10, 12, 14, 15, 16, 17, 18, 19°. Grid used for airfoil in SU2 it has 897 × 257 grid points [39].

Figure 5 shows the lift coefficient of the smooth NACA0012 airfoil as a function of the angle of attack for various mesh sizes.

As can be observed, with increasing mesh size, lift curves exhibited near-coincidence across angles of attack ranging from 0 to 17°. Discrepancies became apparent at higher angles of attack, indicative of flow disruption. For subsequent calculations employing the ESGR model, a third geometry comprising 1793 × 513 cells will be utilized. To facilitate a qualitative assessment of the results, Fig. 6 depicts the flow velocity distributions in the immediate vicinity of the smooth NACA0012 airfoil at an angle of attack of 12°, compared with findings presented in [39].

Employing the ESGR outlined above, identical calculations were performed utilizing a surface roughness value of $k/c = 6e^{-4}$ Fig. 7 presents the lift curves for both the smooth and rough NACA0012 airfoils, plotted in comparison to experimental data and the results obtained from [39].

As illustrated in Fig. 7, surface roughness demonstrably impacts aerodynamic characteristics. Notably, at an angle of attack exceeding 12°, a reduction in lift is observed, correlated with the formation of a vortex at the trailing edge. The resultant vortex is presented in Fig. 8 for comparative analysis alongside findings reported in [39].

To facilitate a qualitative evaluation of the acquired data, Fig. 9 illustrates the spatial distribution of flow velocity and pressure coefficient across the surfaces of both the smooth and rough NACA0012 airfoil.

Analysis of the data presented in Figs. 8 and 9 reveals that the implementation of the roughness model results in a

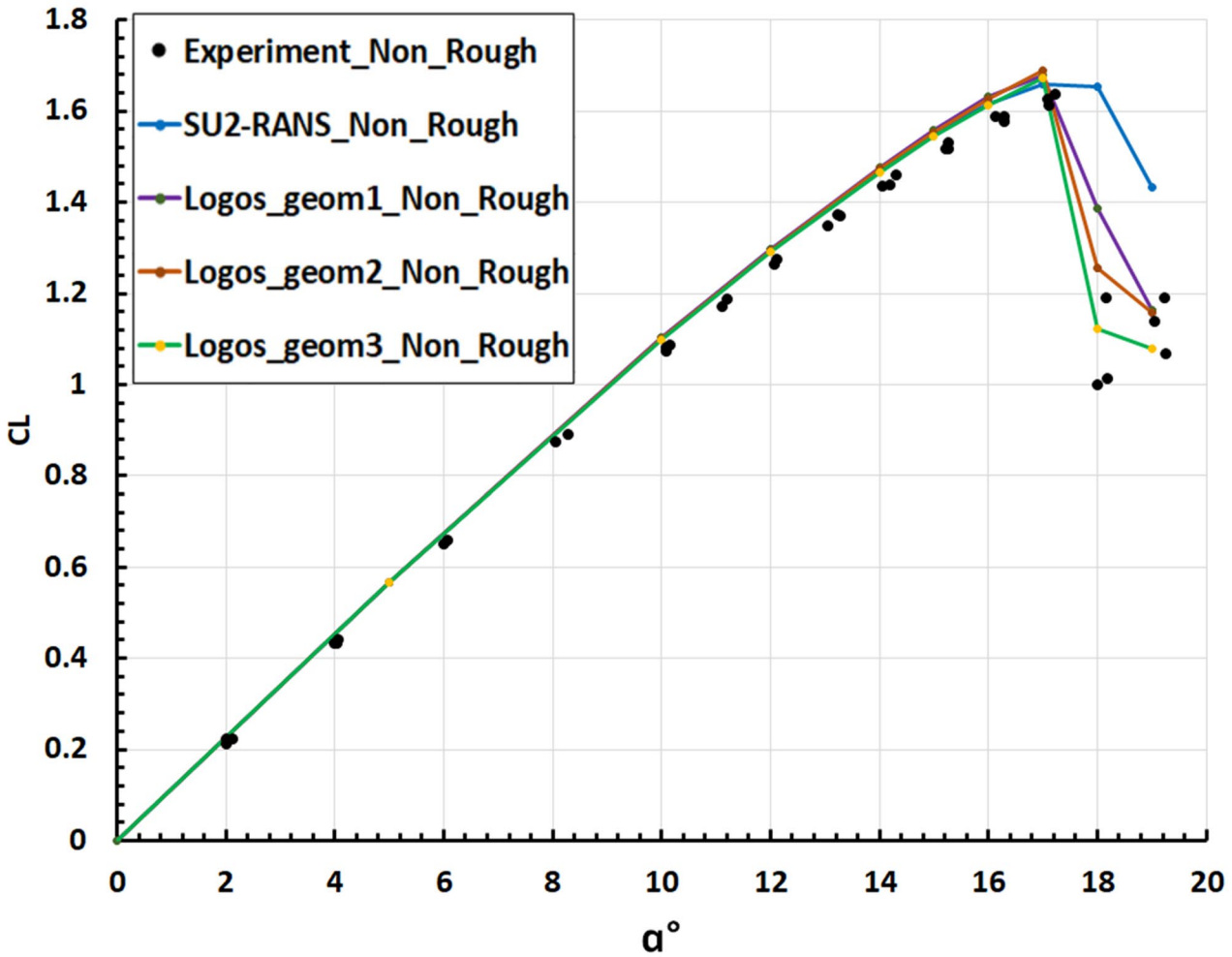


Fig. 5 The lift coefficient of the smooth NACA0012 airfoil as a function of angle of attack

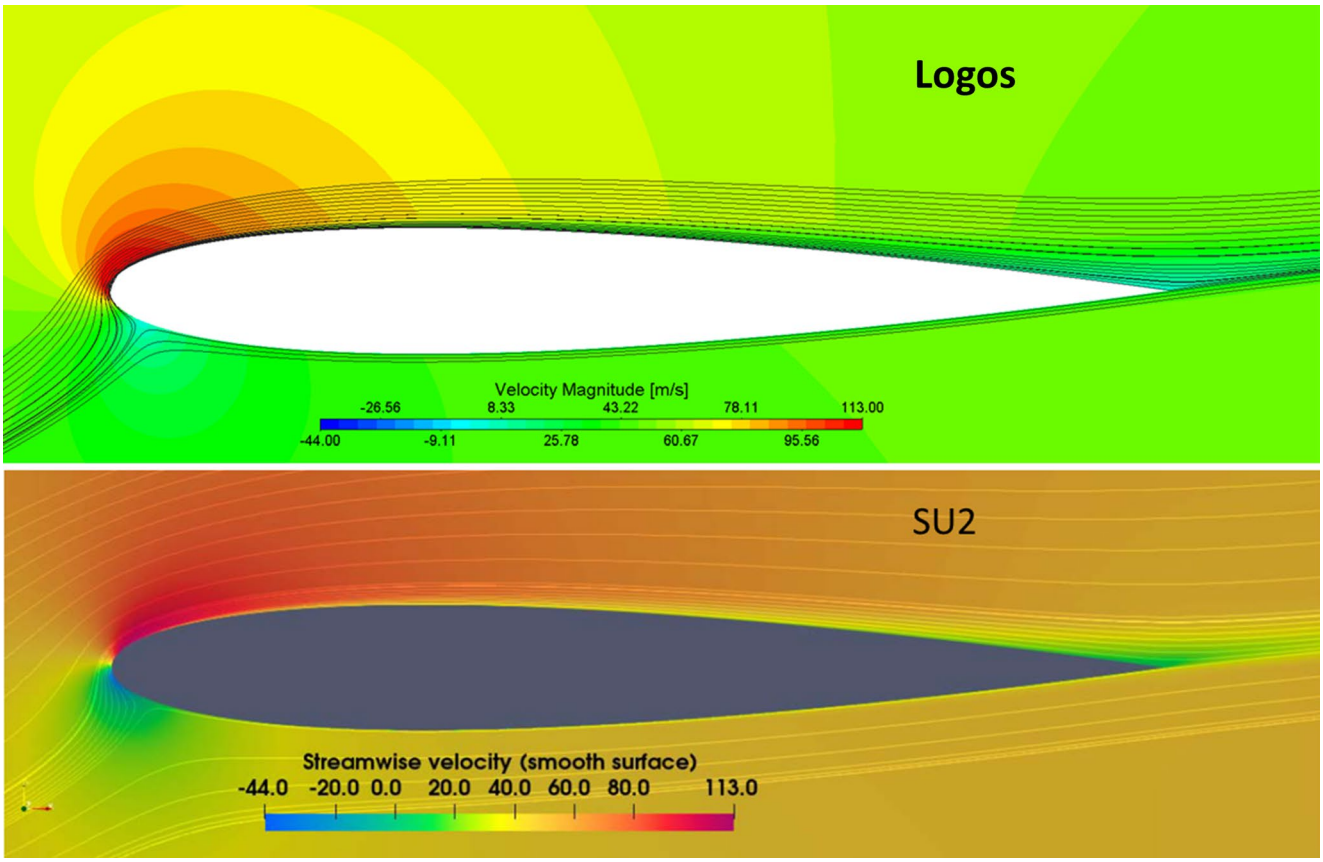


Fig. 6 The velocity fields in the vicinity of the smooth NACA0012 airfoil at an angle of attack of 12°

reduction of the superficial velocity proximal to the airfoil surface. This phenomenon is characterized by the formation of a vortex at the trailing edge of the wing, potentially inducing flow separation and subsequent stall. This effect is described Lynch & Khodadoust [40] and Bragg et al. [41]. As illustrated in Fig. 9, the surface roughness exhibits a negligible impact on the spatial distribution of the pressure coefficient across the airfoil's aerodynamic profile.

3.2 Analysis of the applicability of the ESGR model in solving icing problem

Using the sand roughness model and its modifications presented above, we estimated the influence of various forms of leading edge ice formations on the aerodynamic characteristics of the NACA0012 airfoil. The following parameters were used for numerical studies: the NACA0012 airfoil with a chord length of 0.5334 m, the freestream flow velocity of 58.1 m/s, the freestream pressure of 90,760 Pa, an angle of attack of 3.5°, the water content of 1.3×10^{-3} kg/m^3, an average particle diameter of 20×10^{-6} m, the icing time—8 min [6]. The studies were carried out with the following flow deceleration temperatures [6]: 1) exp. No. 062791.009 (−26.28 °C); 2) exp. No. 062791.008 (−20.28 °C); 3) exp. No. 062791.007 (−18.28 °C); 4) exp. No. 062791.006 (−15.28 °C); 5) exp. No. 062791.005 (−12.28 °C); 6) exp. No. 062791.004 (−8.28 °C); 7) exp. No. 062791.001 (−5.28 °C); 8) exp. No. 062791.002 (−2.28 °C); 9) exp. No. 062791.003 (−1.28 °C).

The calculations were carried out with the following solver parameters: the calculation was performed using the AUSMPW method with gradient reconstruction; the least squares method was used to calculate the gradient; turbulent flows were modeled using the SST turbulence model; wall function: High y+; maximum Courant number: 100.

All problems were calculated in four variants: 1) with a constant value of the equivalent sand-grain roughness height ($k_s = 5e - 4\mu M$); 2) with the NASA correction; 3) with the Shin correction; 4) without roughness. Figures 10, 11, 12, 13 and 14 show the results of calculations taking into

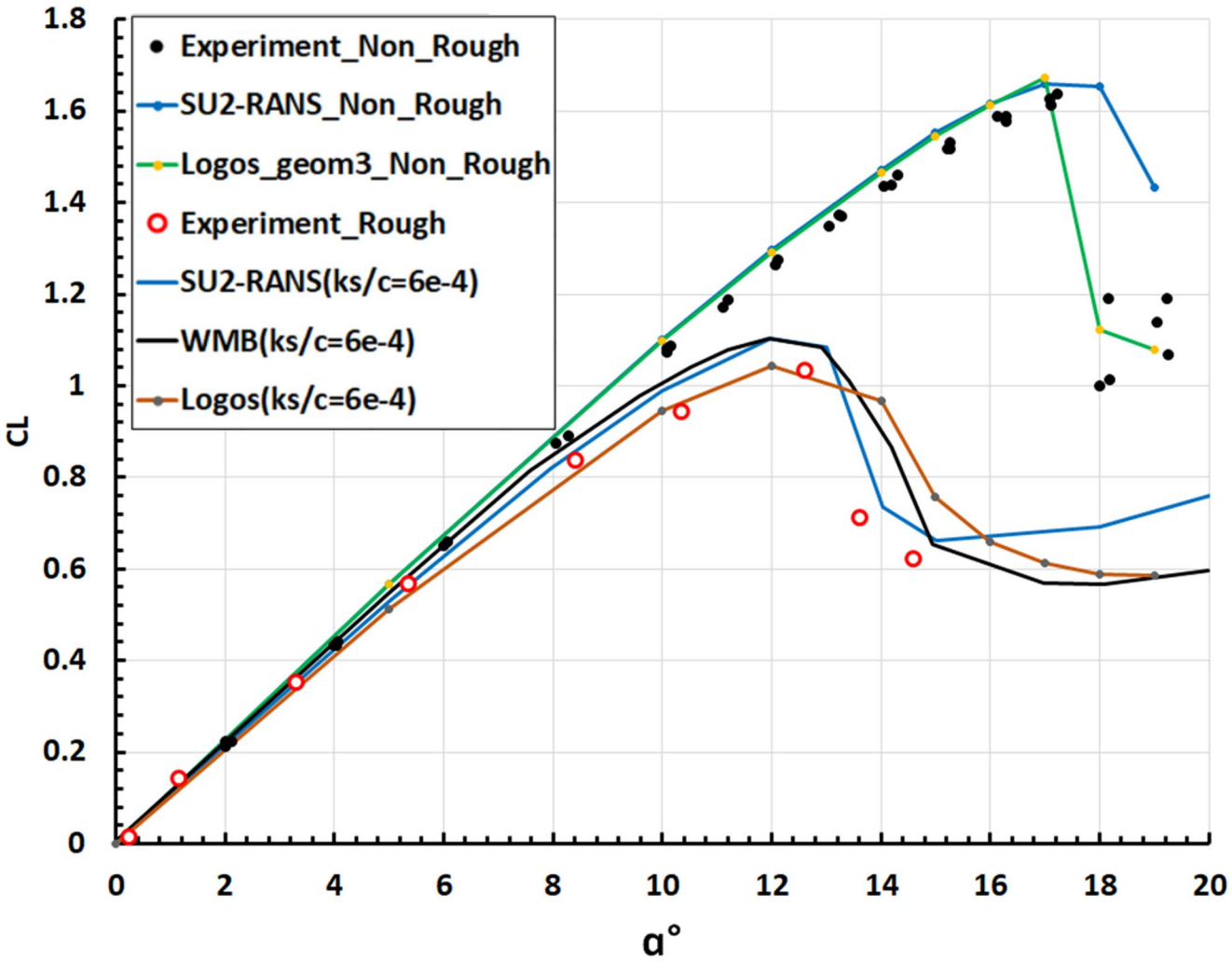


Fig. 7 The lift coefficient of the smooth and rough NACA0012 airfoil as a function of the angle of attack

account roughness in the form of ice formations on the leading edge. For comparison, the figures also show the shape of the clean airfoil, test results [42], as well as the Lewice code results from [6] (where the simulation was made with clean models).

Figure 10 illustrates that for studies 1 and 2, ice accretions primarily form on the leading edge of the airfoil and exhibit a simple geometric shape without horn-like structures. The amount of ice formed on the leading edge in numerical calculations using the Logos is greater than observed experimentally; however, it aligns with results obtained from the Lewice. Notably, applying the roughness model in these tasks has a minimal impact on the resulting outcomes.

Figure 11 illustrates that for tasks 3 and 4, ice accretions primarily form on the leading edge of the airfoil, alongside the formation of small, horn-like structures. Employing the sang grain roughness model alters the position of these horn-like accretions compared to results obtained from the Logos without the model (Logos(D = 20)). Despite discrepancies with experimental data, the obtained results align with the Lewice in terms of the height of ice formation relative to the clean airfoil.

Figure 12 illustrates that without employing a roughness model, the horn-like accretions shift upwards relative to the stagnation point, whereas with the roughness model implemented, the form of the accretions begins to better align with experimental data, differing from the Lewice.

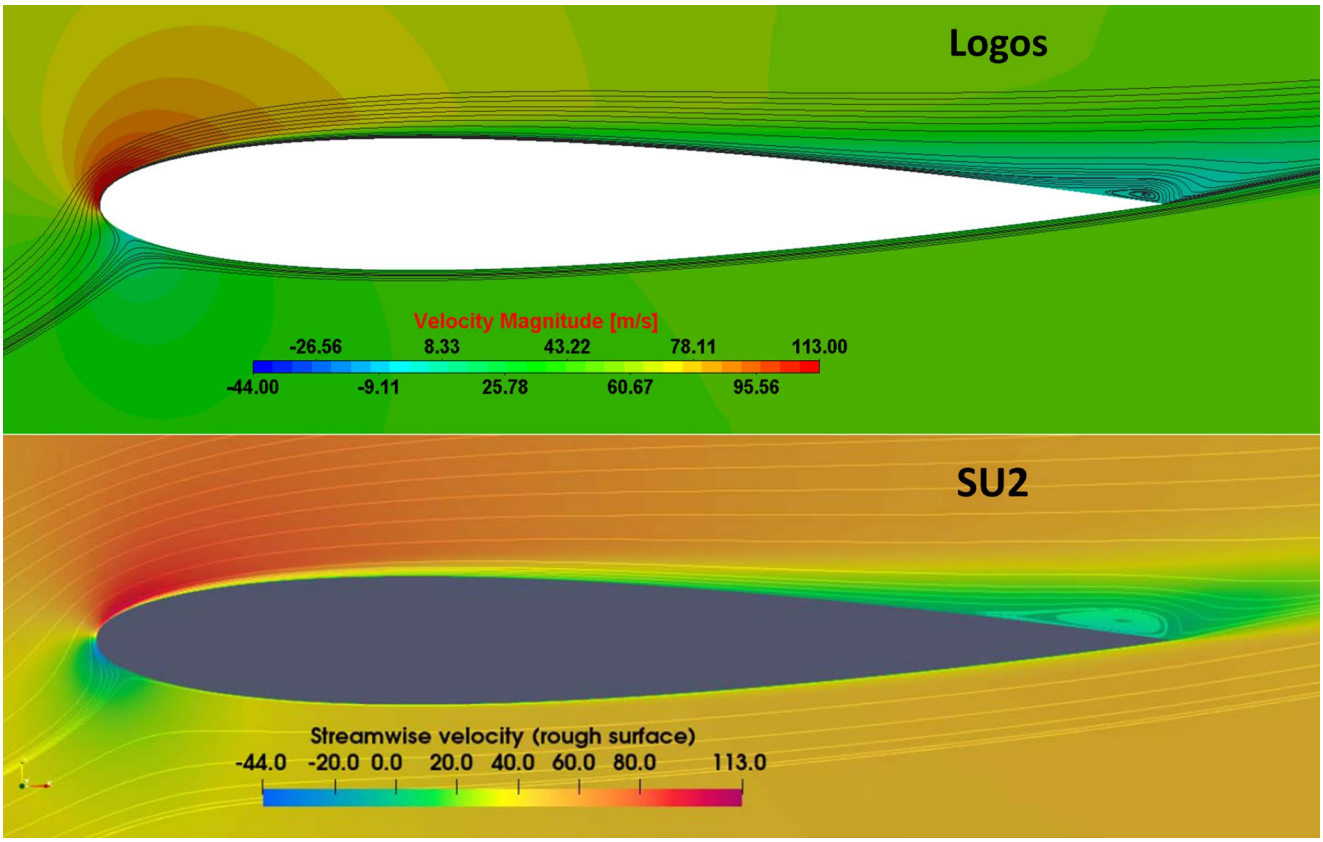


Fig. 8 The velocity fields in the vicinity of the rough NACA0012 airfoil at an angle of attack of 12°

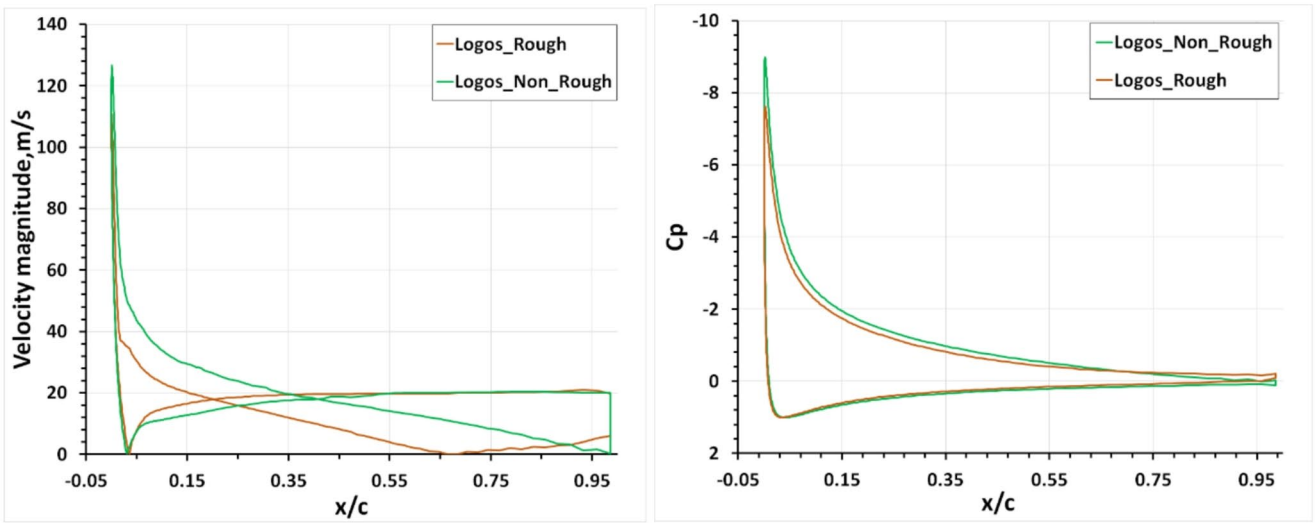


Fig. 9 The velocity and pressure coefficient on the surface of smooth and rough NACA0012 airfoil

Figure 13 demonstrates that without employing a roughness model, the horn-like ice accretions are reduced in size and the ice distribution on the leading edge of the airfoil is more uniform compared to experimental observations. Utilizing the roughness model results in ice accretions with a horn-like morphology, and the location of ice formation on the surface aligns with the experimental data, exhibiting

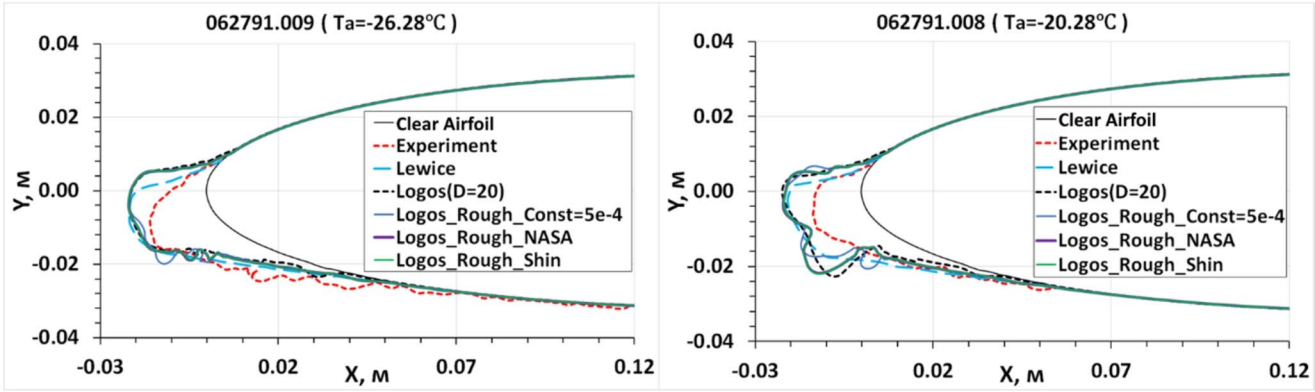


Fig. 10 The NACA0012 airfoil after icing (studies 1 is on the left, studies 2 on the right)

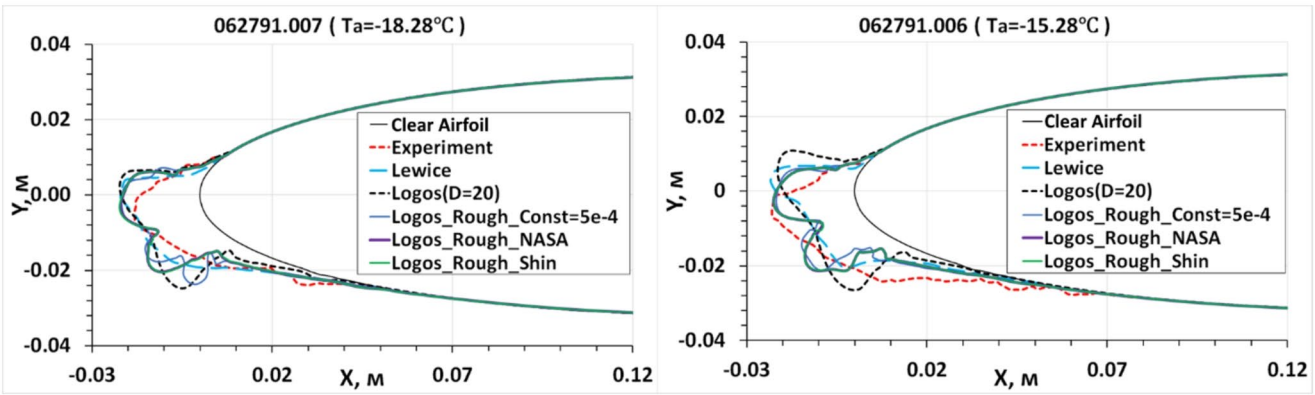


Fig. 11 The NACA0012 airfoil after icing (studies 3—on the left, studies 4—on the right)

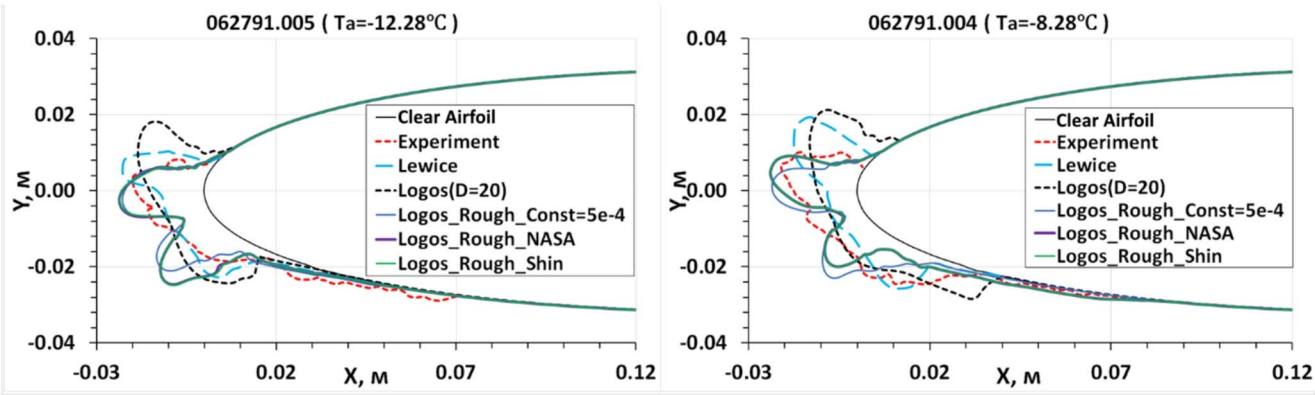


Fig. 12 The NACA0012 airfoil after icing (studies 5—on the left, studies 6—on the right)

a minor discrepancy (approximately 2 mm) in height relative to the experiment.

Figure 14 illustrates that without employing a roughness model, the ice accretions are uniformly distributed across the airfoil profile, differing from experimental observations. The roughness model does not induce spreading of the water film on the surface and, under this icing regime, produces horn-like accretions as observed in the experiment. Consequently, Figs. 10, 11, 12, 13 and 14 demonstrate that the roughness model successfully reproduces the morphology of horn-like accretions, aligning with experimental findings regarding their location and growth orientations – a contrast

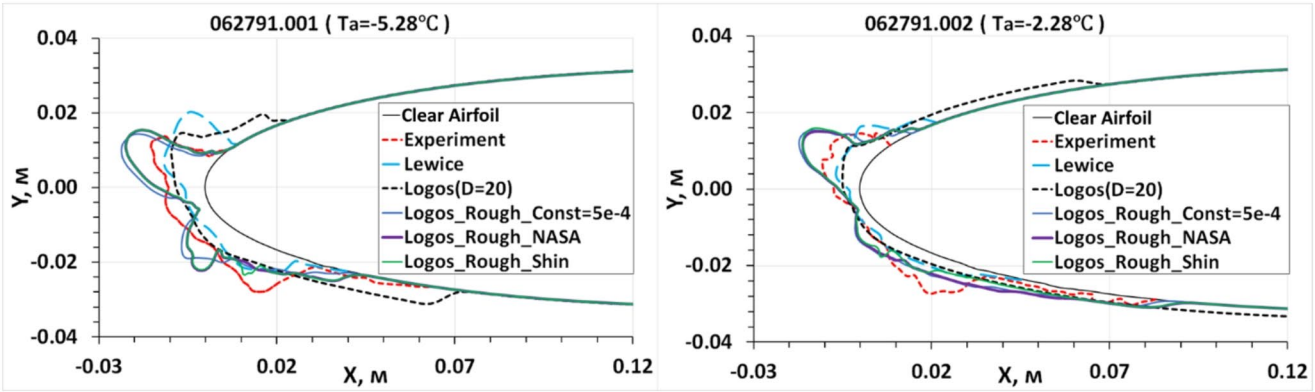


Fig. 13 The NACA0012 airfoil after icing (studies 7—on the left, studies 8—on the right)

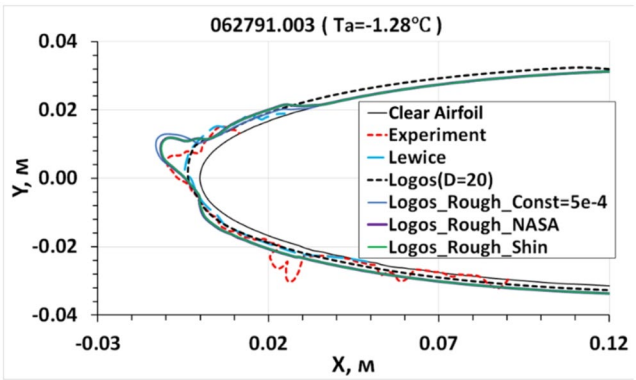


Fig. 14 The NACA0012 airfoil after icing (studies 9)

to results obtained from the Logos and the Lewice without incorporating a roughness model. Agreement with the experimental data is also evident in the distribution of the drag coefficient presented in Fig. 15.

From the shapes of ice formations shown in Figs. 10, 11, 12, 13 and 14 and the graph in Fig. 15, we can conclude that the sand roughness model is able to predict ice formations similar to the experimental ones. This is especially evident in production 6 and 7, where the peak of resistance forces became better consistent with the experiment. The roughness given by the constant, despite the similar pattern of a sharp increase in drag, leads to the formation of ice, which differs from the formations obtained by setting the roughness using NASA and Shin corrections, which can be seen from the underestimated values of drag at a temperature of −5.28 °C.

All calculations were performed using the Russian software package Logos, which is specifically designed for solving computational fluid dynamics problems on arbitrary unstructured grids. The Logos software package underwent rigorous validation [5, 43] and demonstrated favorable performance across a range of hydrodynamic simulations, including turbulent flows and geophysical flow modeling [21, 44].

4 Conclusions

This work presents a physics-mathematical model incorporating a surface roughness model (ESGR) and its modifications from Shin and NASA. Numerical simulations were conducted using this model to investigate the aerodynamics of both smooth and rough NACA0012 airfoil, with and without ice accretion. The results demonstrate that surface roughness influences aerodynamic characteristics; specifically, stall—accompanied by a reduction in lift—occurs at lower angles of attack on the rough profile compared to the smooth profile. Vortex formation can also occur at lower incoming flow velocities, as evidenced by differences in the structure and behavior of the flow over rough versus smooth surfaces, consistent with findings reported in [11, 14]. When utilizing the ESGR model for icing simulations, it was shown that the shapes and locations of ice accretion more closely align with experimental data than those predicted for a smooth surface. This is attributed to reduced water film spreading on the rough surface, resulting in “horn” accretions observed in experimental studies [50]. A key feature of this work is the comprehensive set of results obtained using the roughness model across various icing temperature regimes applied to the NACA0012 airfoil. These results reveal a correlation indicating that incorporating the roughness model at incoming flow temperatures above −15 °C yields accretion profiles more consistent with experimental findings. Despite remaining discrepancies, the presented results provide direction for further research into other factors influencing icing.

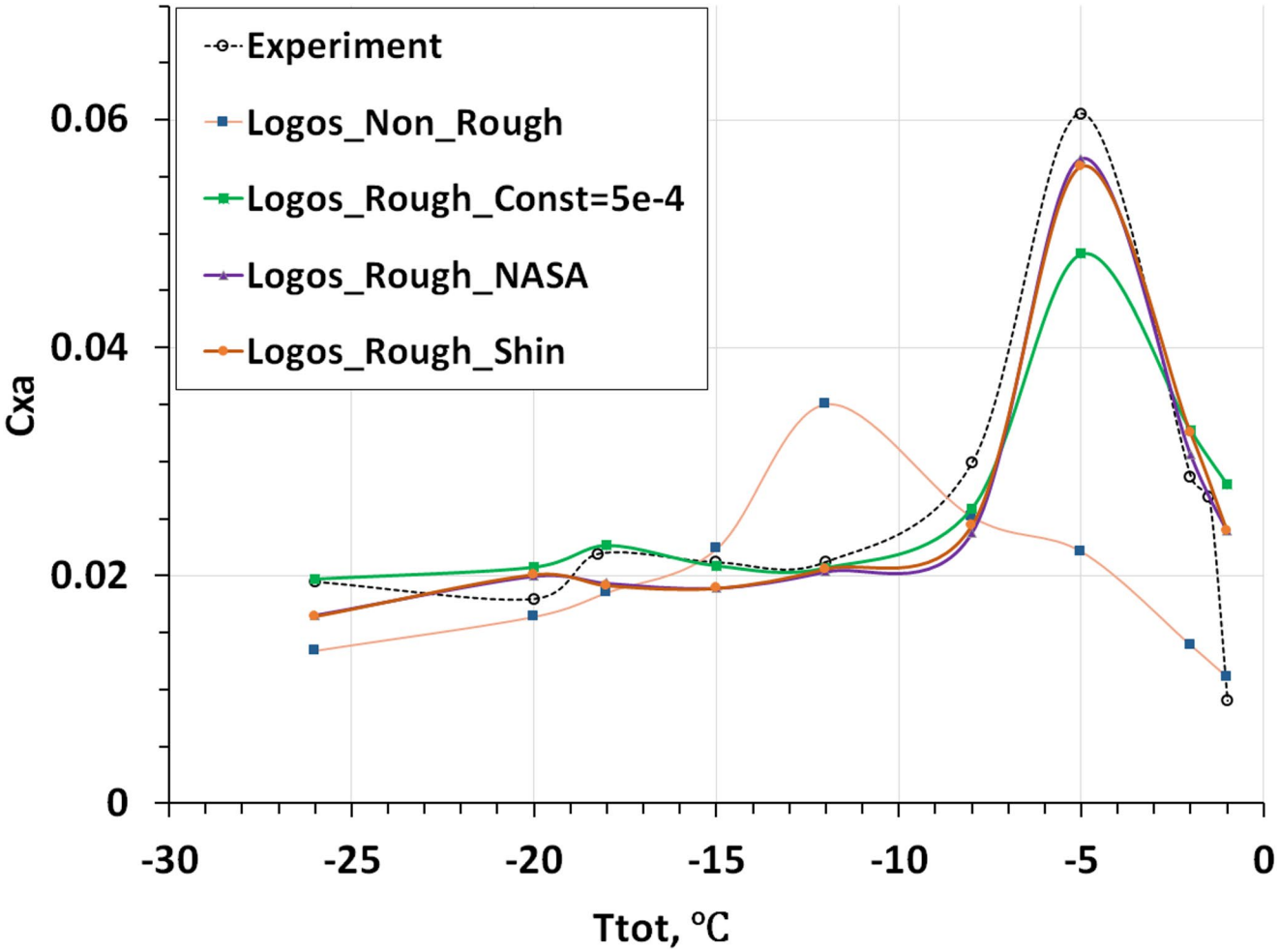


Fig. 15 The drag coefficient of the NACA0012 airfoil as a function of the total temperature

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Code availability Not applicable.

Declarations

Conflict of interest Not applicable.

Competing interests The authors have no competing interests to declare that are relevant to the content of this article.

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